De novo computational design of epitope-specific nanobodies against the *Orthobunyavirus oropoucheense* nucleocapsid for point-of-care diagnostics

Alcántara-Ríos, C.[1] ; Hernandez, A.[1] ; Sandoval, G.[1]
[1] Faculty of Biological Sciences, Universidad Nacional Mayor de San Marcos, Peru

Abstract

In this work, we developed an epitope-directed computational workflow for the accelerated design of nanobodies against the Oropouche complex. Two highly conserved, pan-specific epitopes were identified: Epitope B (residues 75–85) and Epitope C (residues 171–186), both without cross-reactivity with Zika, Dengue, Chikungunya, and Mayaro viruses. Of the 96 designs generated, 14 passed the initial structural filters. The candidates for Epitope C met stringent quality metrics (pLDDT ≥ 0.75, pTM ≥ 0.70, interface RMSD up to 1.03 Å), stabilized by 3 hydrogen bonds and 3 salt bridges according to PDBePISA. Epitope B, on the other hand, exhibited greater flexibility and was retained under more relaxed criteria. Molecular dynamics simulations for 100 ns revealed that, although the protein backbone underwent conformational sampling (RMSD of 7.98–8.32 Å), the interfacial hydrogen bond network remained highly stable, averaging more than 7.6 bonds per frame in both systems. This study validates a generalizable and low-cost framework for the design of targeted nanobodies, advancing the development of point-of-care diagnostics in resource-limited regions.

Background

The 2023-2025 Oropouche outbreak, with over 23,000 cases across the American continent, exposed critical public health gaps. Although RT-qPCR remains the diagnostic gold standard, it requires specialized infrastructure inaccessible in endemic rural areas. Coupled with reinfection rates approaching 60%, these gaps underscore the urgent need for low-cost point-of-care (POC) alternatives such as lateral flow assays. Traditional nanobody discovery relies on animal immunization, which is slow and unable to target predefined epitopes, and no epitope-directed POC reagents have been reported for the Orthobunyavirus oropoucheense species complex.

Methodology

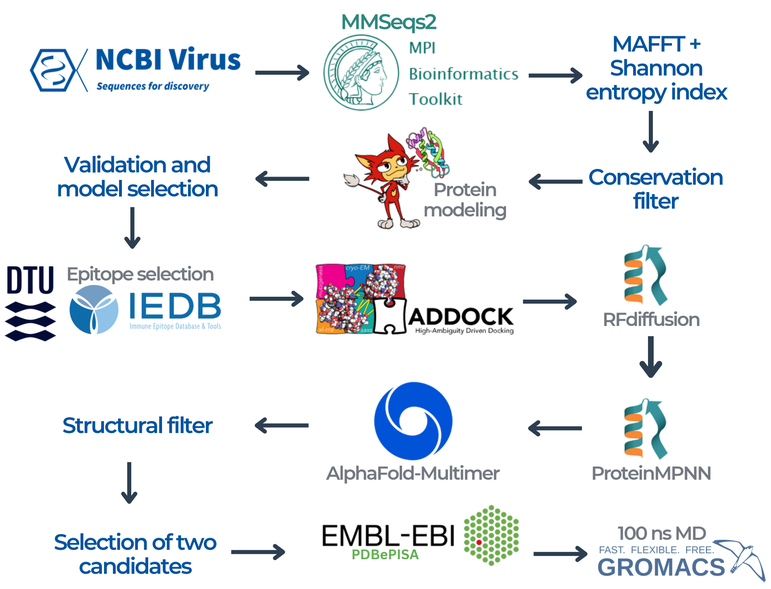


Fig 1. Pipeline used to design and validate clusters.

Complex formation



Fig. 4. Interaction between *de novo* nanobodies (cyan and green) and predicted epitopes on the OROV N protein (magenta and blue), forming the basis for a lateral flow assay design.

Results

Two Pan-Specific Epitopes Identified

Epitope B
- Shannon entropy: 0.01290 (highly conserved)
- Moderate RNA-binding interference
- SASA: 23.99-117.93 Å² per residue

Epitope C
- Shannon entropy: 0.01667 (highly conserved)
- Low RNA-binding interference (exposed epitope)
- SASA: 0.80-170.70 Å² per residue

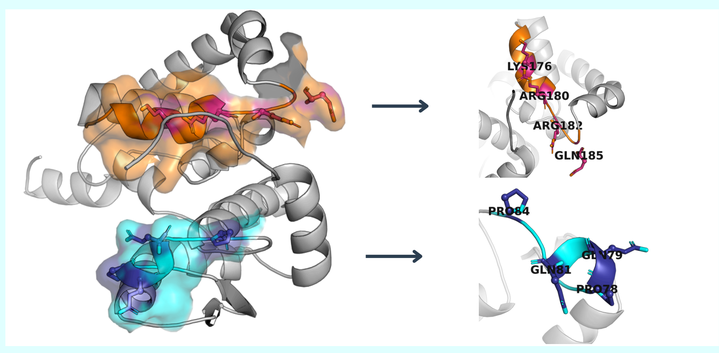


Fig 2. Selected target epitopes, Epitope B (cyan) and Epitope C (orange), with highlighted hotspot residues (blue and magenta).

Selection funnel

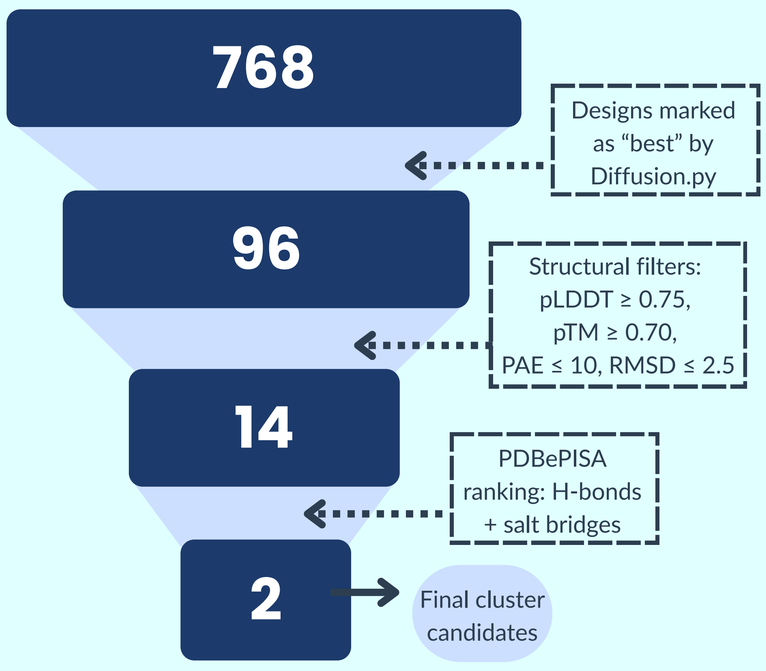
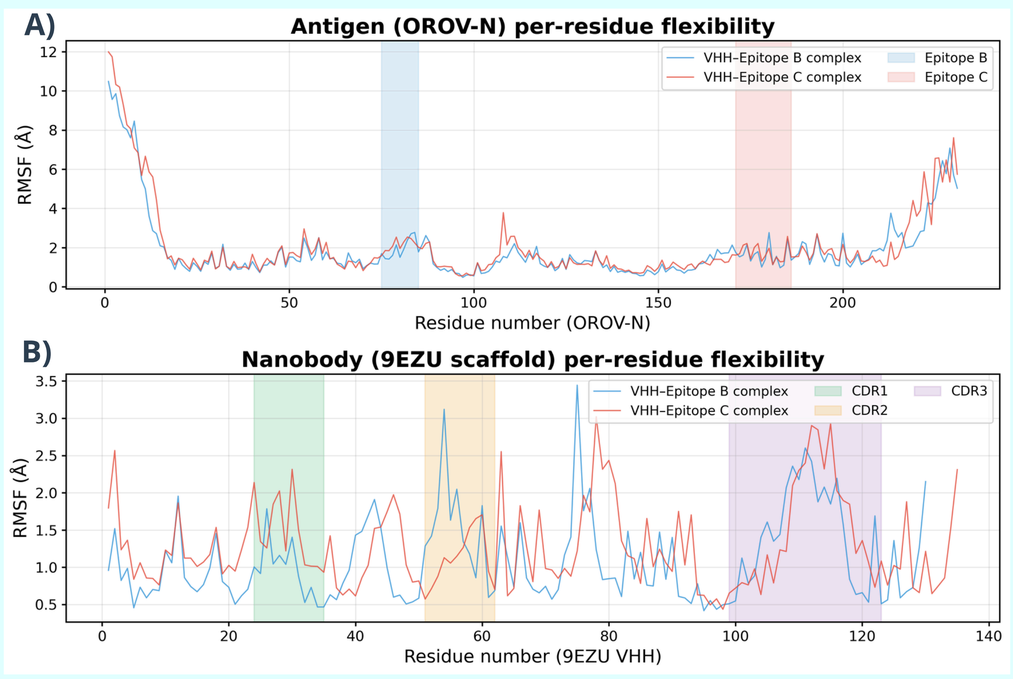
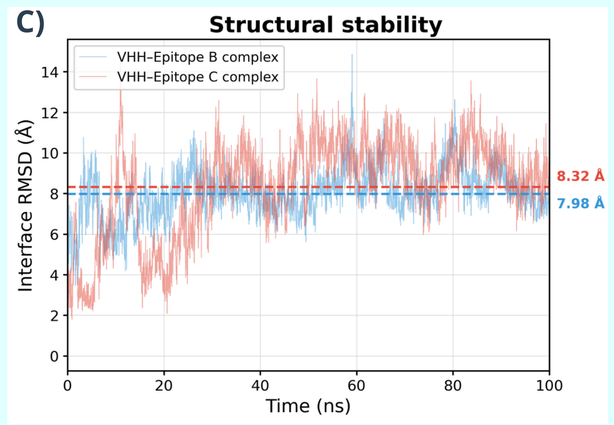


Fig. 3. Selection funnel. Epitope B clusters were retained under relaxed criteria (pLDDT ≥ 0.65, pTM ≥ 0.55, PAE ≤ 15, RMSD ≤ 5.0).

100 ns molecular dynamics simulations

- RSMD reflects conformational sampling typical of *de novo* designs
- Persistent H-bond network predicts engaged binding in the epitope-nanobody cluster interface despite global fluctuations





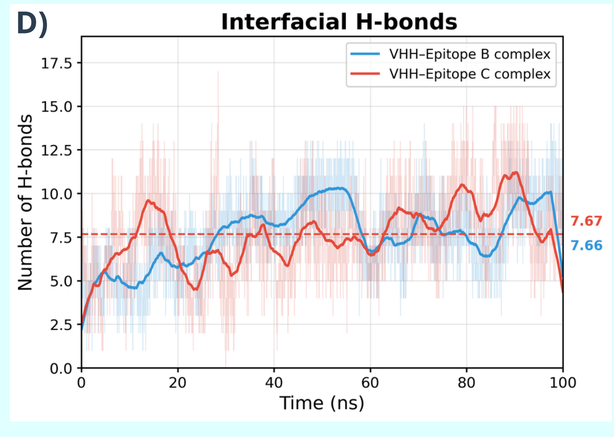


Fig. 5. Molecular dynamics analysis. (A) RMSF of OROV-N across complexes. (B) RMSF of the designed nanobodies. (C) RMSD profiles of the VHH–epitope complexes. (D) Number of intermolecular hydrogen bonds at the binding interface throughout the 100 ns simulation.

Conclusions

- This workflow provides a generalizable, epitope-directed framework for rapid, low-cost nanobody design against emerging arboviruses, supporting the accelerated development of species-specific POC lateral flow assays for resource-limited settings in Latin America.
- Successfully designed 2 epitope-specific nanobodies *de novo* showing high sequence conservation across the Oropouche complex and no cross-reactivity with co-circulating arboviruses (Zika, Dengue, Chikungunya, and Mayaro).
- Future directions include the experimental validation (in vitro) of these designed nanobodies and the expansion of the screening library to optimize binding affinities for point-of-care diagnostic integration

Bibliography

Acknowledgements

To our professors at the National University of San Marcos, especially Dr. Erasmo Colona and Dr. Libertad Alzamora, for their support and interest in guiding students toward new frontiers of science; and to Luis Rodríguez Carnero for his advice and expertise in Immunology.